Speaker script: CLL798D final presentation

District heating pipe network layout using MSTs · 6 minutes + 5 minutes Q&A · about 800 words, read at a normal pace

Slide 1: Title

0:00 – 0:10

Good afternoon. Our project is on designing the pipe layout of a district heating network using graph algorithms, mainly minimum spanning trees. Today we will only talk about the work we did after the prelim.

Slide 2: Prelim, and what we did after it

0:10 – 0:50

On the left is what we showed at the prelim. We built MST layouts for two real German districts, Bremen which is flat and Stuttgart which is hilly. The terrain-weighted MST was 3 percent cheaper in Stuttgart, Stuttgart needed two pressure zones, and one extra pipe removed 57 percent of the unserved demand. The weak point was that all of this came from just one network per city.

On the right are the five things we did after the prelim. Points 4 and 5 are the two next steps we had promised at the prelim. I will go through them one by one.

Point to the left column, then the right column.

Slide 3: 1. The 3% terrain saving changes a lot

0:50 – 1:40

First, we checked if our 3 percent result was reliable. We used the same two districts, but built the network in many different ways: 6 different numbers of substations and 10 different k-means clusterings each, so 120 networks in total.

Each dot here is one network, and the circled dot is our prelim result. For Stuttgart with the same 40 substations, the saving goes anywhere from minus 6 to plus 13 percent, just because the substations moved. Bremen stays at zero, as expected for flat ground.

So the idea is right on average, but the exact 3 percent was partly luck. The reason is that the MST only looks at edge lengths. The real cost depends on how much flow each pipe carries, and the MST cannot see that.

Point to the circled dot, then to the spread of orange dots above 40.

Slide 4: 2. Edge swaps give a cheaper tree

1:40 – 2:40

So we tried to improve the MST using the actual cost instead of length. The method is simple. We start from the MST, add one nearby edge, which creates a cycle, and remove another edge from that cycle, so we again have a tree. If the full cost model says it is cheaper, we keep the change. We repeat until nothing improves.

On the left is the MST and on the right is the result. The orange edges are the ones that changed. The new tree is about 3 percent longer, but 9 percent cheaper.

We tested this on 20 networks and it was cheaper in all of them. It works in flat Bremen too. So our prelim statement that the MST is the best answer on flat ground was not correct.

Point to the orange edges near the star (the plant).

Slide 5: 3. Cost depends on distance to the plant

2:40 – 3:30

The obvious question is: why is a longer tree cheaper? To answer this we took 30 different trees and plotted their cost.

On the left, cost against total pipe length. There is no clear relation. On the right, cost against the average distance from each substation to the plant, weighted by its heat demand. This is almost a straight line, with correlation 0.98.

So what matters is how far the heat has to travel, not how much pipe we lay. Long paths need bigger pipes and more pumping. In graph terms, the MST minimises total length, a shortest-path tree minimises path length, and the cheapest network is somewhere in between.

Left plot first, then right plot.

Slide 6: 4. Pressure zones switch at 151 m

3:30 – 4:20

Next, pressure zones. At the prelim we showed that one zone with PN16 pipes can handle at most 151 metres of height difference, so Stuttgart needs two zones. We had planned to test this on a third city, but we could not get the data.

So instead we took Stuttgart's terrain and scaled the heights, from completely flat up to 1.5 times the real height. Blue dots need one zone, orange dots need two. The switch happens exactly at 151 metres, as predicted.

The black line also shows that the terrain saving is real: it is zero when flat and grows as the ground gets steeper.

Point to the vertical 151 m line, then the black line.

Slide 7: 5. Greedy redundancy vs exact ILP

4:20 – 5:10

Last, reliability. A tree has no backup, so if one pipe breaks, everything after it loses heat. At the prelim we added backup pipes using a greedy method, and we said we would check it against an exact optimisation.

We wrote it as an integer linear program: pick the extra pipes that protect the most demand, with the same budget as greedy. The black line is the exact optimum, the dots are greedy, for 1 to 8 extra pipes. Greedy is never more than 1.5 points below. So greedy is good enough.

And if we combine both ideas, the edge-swap tree plus one backup pipe is as reliable as the MST plus one backup pipe, and about 5 percent cheaper.

Point to where the dots sit on the line.

Slide 8: Summary

5:10 – 5:45

To summarise. The terrain saving is real on average, but noisy for a single network. Our claim about flat ground was wrong. The pressure zone result and the greedy redundancy both hold up.

The main point is: the MST is a good starting point, but the cheapest tree is a little longer and has shorter paths to the plant. Thank you, we are happy to take questions.

Stop here. Backup slides are only for questions.

Backup slides (only if asked)

Slide 9: Backup: seasonal load model

Q&A; only

If asked about costs: at the prelim we assumed pumps run at full power for 5,000 hours a year. In reality heating demand changes with the season, so we used a standard load-duration curve. Pump energy came out about 8 times lower. The curve is a textbook formula, not measured data, which is one of our limitations.

Slide 10: Backup: sensitivity and network size

Q&A; only

If asked about prices or network size: changing electricity price, heat price, discount rate or supply temperature changes the numbers but never changes which layout is cheapest. The edge-swap gain also holds for bigger networks: around 8 to 9 percent with 80 and 160 substations.

Likely questions

Q. Where did the data for the 120 networks come from?

It is the same data as the prelim: the AixDHN dataset from RWTH Aachen (heat demand per census cell) and NASA SRTM 30 m elevation. We cluster the cells into substations with k-means. We repeated that with 6 substation counts (20 to 160) and 10 random seeds each, for both cities. So these are 120 ways of dividing the same two districts, not 120 new districts.

Q. Why does the clustering change the result so much?

Substation positions are our modelling choice. Moving them changes which edges the MST picks, and that changes the flow paths. The terrain MST changed only 3 of 40 edges, but those edges happened to shorten the main flow path, which cut pumping cost.

Q. Is the edge-swap tree the optimum?

No, it is a local search, so there is no guarantee. But starting from two different MSTs gives almost the same final cost (within 0.09 EUR/MWh), which suggests it is not stuck in a poor local minimum.

Q. Why not just use a shortest-path tree or a star?

Those minimise path length but use much more pipe. The star was the most expensive layout. The cheapest tree is between the MST and the shortest-path tree, which is what the edge swaps find.

Q. What exactly is the ILP?

One binary variable per candidate extra pipe and one per tree edge. An edge counts as protected if at least one chosen pipe closes a cycle through it. We maximise the protected unserved demand, with the same number of pipes and the same budget as greedy. Solved with HiGHS through scipy in under a second.

Q. Why scale the terrain instead of using a third city?

We could not get the elevation data for a third district in time. Scaling one city's terrain is a controlled test: only the height changes and everything else stays fixed. A real third district would still be a better test, and we list it as a limitation.

Q. Where does the 151 m limit come from?

Pipe rating 16 bar, minus the vapour pressure of water at 110 °C, minus a 0.5 bar safety margin, divided by ρg. That gives about 151 m of water column.

Q. How realistic is the seasonal load curve?

It is a standard textbook formula (Sochinsky) fitted to 1,800 full-load hours. We do not have hourly data for these districts. We changed the summer base load from 5 to 20 percent and the conclusions did not change.